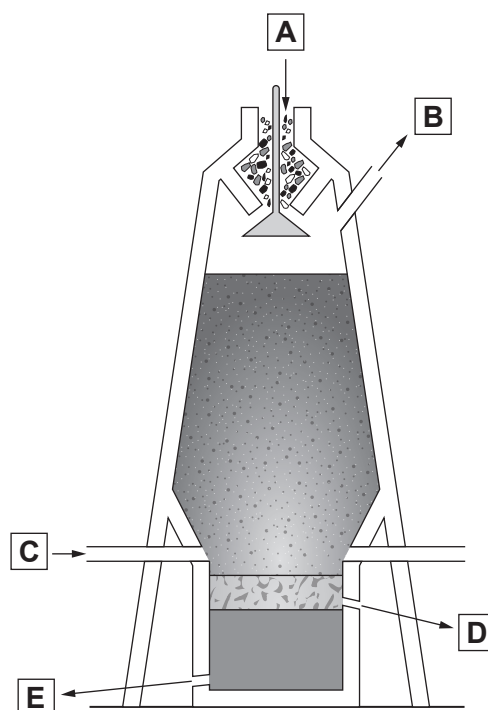


Answer **all** questions.

1. (a) The diagram shows a blast furnace which is used to extract iron.

Labels **A**, **B**, **C**, **D** and **E** show where substances enter and leave the furnace.



- (i) Give the letter, **A**, **B**, **C**, **D** or **E**, that shows where

iron ore enters the furnace

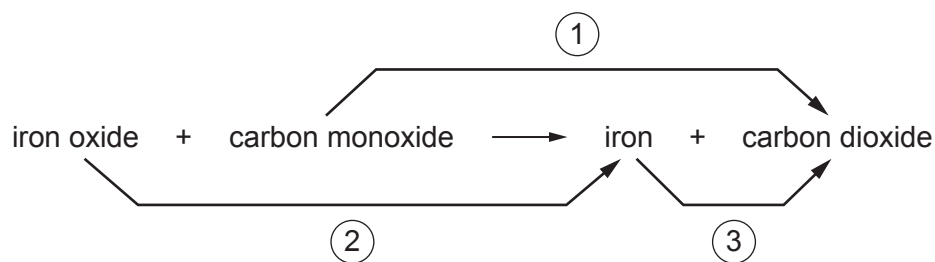
waste gases leave the furnace

slag is removed

[3]



- (ii) One of the main reactions inside the blast furnace is represented by the following equation.



Give the number of the arrow that shows oxidation.

[1]

.....

- (iii) Limestone is added to the furnace to remove impurities.

The chemical name for limestone is calcium carbonate.

Calcium carbonate contains the ions Ca^{2+} and CO_3^{2-} .

Circle the correct formula for calcium carbonate.

[1]

CaCO**Ca₂CO₃****CaCO₃****Ca(CO₃)₂**

- (iv) One of the waste gases that leaves the blast furnace is carbon dioxide, CO_2 .

Put a tick (✓) in the box next to the diagram that best represents a carbon dioxide molecule.

[1]

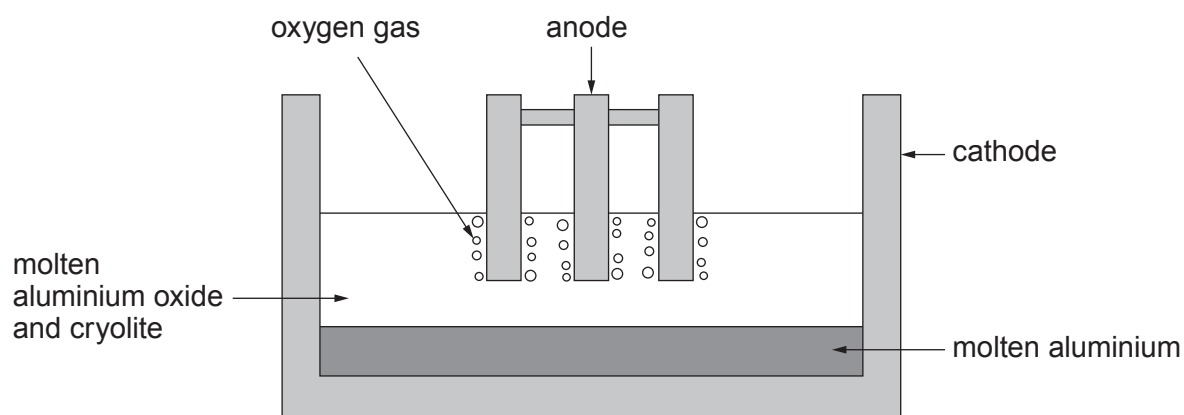

☐

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(b) The diagram shows an electrolysis cell used in the extraction of aluminium.



(i) Draw **one** line from each term to its correct definition.

[3]

Term	Definition
anode	positive electrode
electrolyte	a substance that removes impurities
electrolysis	a substance that is split up during the process
	using electricity to make a compound
	using electricity to split up a compound
	negative electrode



- (ii) Underline the correct word in the brackets to complete each sentence. [3]

Cryolite is added to lower the (**density** / **melting point** / **boiling point**) of the electrolyte.

When choosing a location for an aluminium plant in the UK, it is important to be near a port to (**import** / **export** / **clean**) the aluminium ore.

At the temperature inside the cell, the aluminium is produced as a (**solid** / **liquid** / **gas**).

- (iii) The equation for the reaction that takes place during the extraction of aluminium is given below.



Choose a number from the box to balance the equation.

[1]

2	4	6
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3430U501
05

13

